Introduction to Probability and Statistics

(EE 354 / CE 361)

Bayes' Rule and Independence

Today's Agenda

- Recall: Bayes' Rule
- Independence
- Conditioning and Independence
- The Tale of a Biased Coin

Bayes' Rule: Example

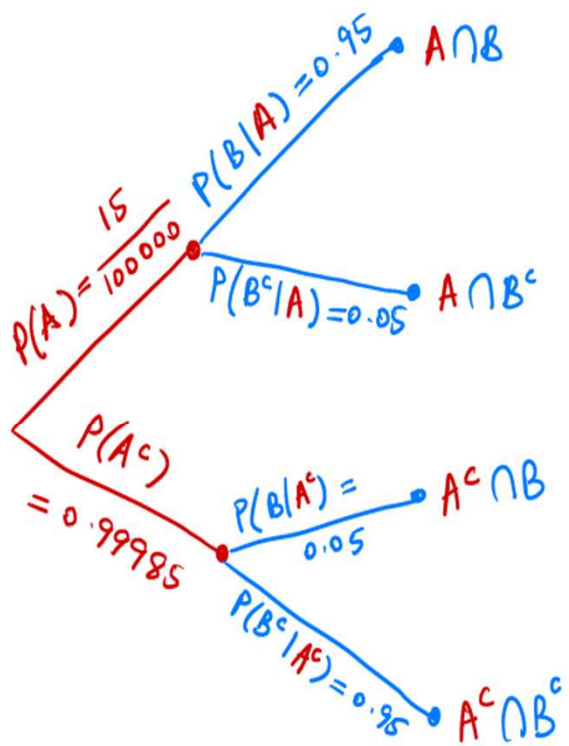
The incidence rate of a certain disease is 15/100000. There is a test for the disease which is 95% accurate(i.e. If a person has the disease, the test comes back positive with probability 0.95. If a person does not have the disease, it comes back negative with probability 0.95). Given that a person tested positive, what is the probability they have the disease.

Instructions:

1. Make a sequential tree diagram. Label all the branches
2. Find the probabilities of each branch
3. Find the probability that a person has the disease, given they tested positive

$A = \{\text{Have the disease}\}$

$B = \{\text{Testing positive}\}$



$$P(A|B) = \frac{P(A) \cdot P(B|A)}{P(A) \cdot P(B|A) + P(A^c) \cdot P(B|A^c)}$$

$$= \frac{0.00015 \times 0.95}{0.00015 \times 0.95 + 0.99985 \times 0.05}$$

$$= 0.00284 \quad (\sim 0.3\%)$$

Independence

- With Conditional Probability, we encode our **updated belief** about the likelihood of an event **IF** we have additional partial information about the results of the random experiment.
- Consider a special case where the occurrence of an event B provides no such additional information about the preceding A i.e.,

$$\mathbf{P}(A|B) = \mathbf{P}(A)$$

In such cases, we say A is independent of B

Independence

$$\mathbf{P}(A|B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

$$\mathbf{P}(A|B) = \mathbf{P}(A)$$

$$\mathbf{P}(A) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

$$\boxed{\mathbf{P}(A \cap B) = \mathbf{P}(A).\mathbf{P}(B)}$$

Independence

$$\boxed{\mathbf{P}(A \cap B) = \mathbf{P}(A).\mathbf{P}(B)}$$

- Why is this definition better than: $\mathbf{P}(A) = \mathbf{P}(A|B)$
- It can be used when $\mathbf{P}(B) = 0$. In such situations, $\mathbf{P}(A|B)$ is undefined

Independence: Example

Example 1.19. Consider an experiment involving two successive rolls of a 4-sided die in which all 16 possible outcomes are equally likely and have probability 1/16.

Are the following events independent?

A = {1st roll results in 1} B = {2nd roll results in 3}

Independence: Example

Example 1.19. Consider an experiment involving two successive rolls of a 4-sided die in which all 16 possible outcomes are equally likely and have probability 1/16.

Are the following events independent?

A = {1st roll results in 1} B = {2nd roll results in 3}

$\Omega = \{(1,1), (1,2), (1,3), (1,4),$
$(2,1), (2,2), (2,3), (2,4),$
$(3,1), (3,2), (3,3), (3,4),$
$(4,1), (4,2), (4,3), (4,4)\}$

$A = \{(1,1), (1,2), (1,3), (1,4)\}$

$B = \{(1,3), (2,3), (3,3), (4,3)\}$

$A \cap B = \{(1,3)\}$

$P(A \cap B) = 1/16$

$P(A) = \frac{4}{16} = \frac{1}{4}$

$P(B) = \frac{4}{16} = \frac{1}{4}$

$P(A \cap B) = P(A) \cdot P(B)$

$1/16 = 1/4 \cdot 1/4 = 1/16$ ✓

Events A and B are independent.

Independence is Symmetric

- If A is independent of B, B is independent of A
- We say that A and B are independent events
- Therefore the statement $\mathbf{P}(A|B) = \mathbf{P}(A)$ implies:

Independence: Example

Example 1.19. Consider an experiment involving two successive rolls of a 4-sided die in which all 16 possible outcomes are equally likely and have probability 1/16.

Are the following events independent?

A = {Maximum of two rolls is 2} B = {Minimum of two rolls is 2}

Independence: Example

Example 1.19. Consider an experiment involving two successive rolls of a 4-sided die in which all 16 possible outcomes are equally likely and have probability 1/16.

Are the following events independent?

A = {Maximum of two rolls is 2} B = {Minimum of two rolls is 2}

$\Omega = \{(1,1),(1,2),(1,3),(1,4),$
$(2,1),(2,2),(2,3),(2,4),$
$(3,1),(3,2),(3,3),(3,4),$
$(4,1),(4,2),(4,3),(4,4)\}$

$A = \{(1,2),(2,2),(2,1)\}$

$B = \{(2,2),(3,2),(4,2),(2,3),$
$(2,4)\}$

$P(A) = \frac{3}{16}$ $P(B) = \frac{5}{16}$

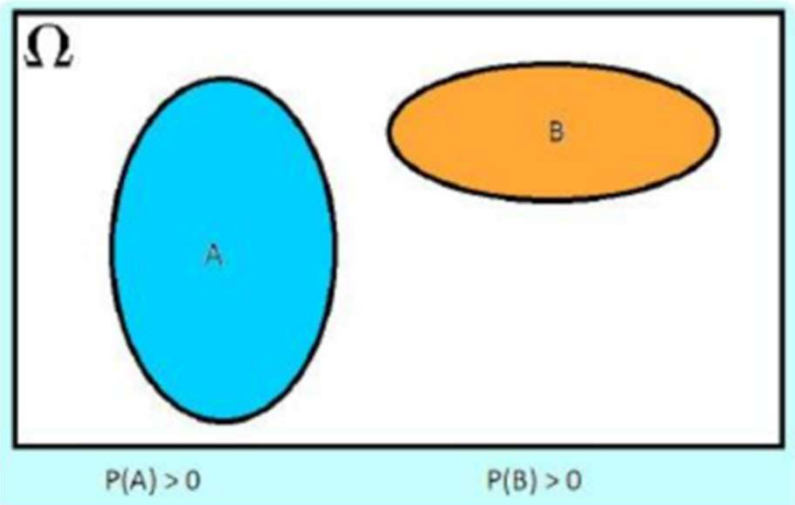
$A \cap B = \{(2,2)\}$, $P(A \cap B) = \frac{1}{16}$

$P(A \cap B) \neq P(A) \cdot P(B)$

$\frac{1}{16} \neq \frac{3}{16} \cdot \frac{5}{16} \Rightarrow \frac{1}{16} \neq \frac{15}{256}$

✗ A & B are NOT INDEPENDENT

Are Two Disjoint Events Independent?



$P(A \cap B) = 0$

$P(A), P(B) > 0$

$P(A \cap B) \neq P(A) \cdot P(B)$

Disjoint Events with positive probability are ALWAYS DEPENDENT on one another.

NOTE: If $P(A)$ or $P(B) = 0$, then they can be independent

Independence: Modelling Application

You are trying to Model "How different factors affect student academic Performance."

$A = \{$ Student has GPA $\geq 3.8\}$

$B = \{$ Student gets ≥ 8 hrs of sleep $\}$

$P(A \cap B) \neq P(A) \cdot P(B)$ → Implies that A & B are dependent on one another.

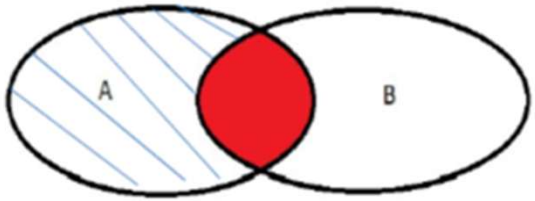
$C = \{$ Student's grandmother has > 15 years of formal education $\}$

$P(A \cap C) = P(A) \cdot P(C)$ → Implies that A & C are independent

Independence of Event Complements

If A and B are independent, then A and B^C are also independent

- Intuitive argument:
  - If A and B are independent, the occurrence of B does not provide any new information on the probability of A occurring.
  - It is then intuitive that non-occurrence of B (i.e. B^C) should also provide no information on the probability of A having occurred.



$$A = (A \cap B) \cup (A \cap B^c)$$

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

$$P(A) = P(A) \cdot P(B) + P(A \cap B^c)$$

$$P(A \cap B^c) = P(A)[1 - P(B)]$$

$$P(A \cap B^c) = P(A) \cdot P(B^c)$$

Conditional Independence

- Conditional Independence, given an event C, is defined as independence under the probability law $\mathbf{P}(\cdot|C)$
- Given a (conditioning) event C, the events A and B are called conditionally independent if:

$$\boxed{\mathbf{P}(A \cap B|C) = \mathbf{P}(A|C).\mathbf{P}(B|C)}$$

- Given a (conditioning) event C, the events A and B are called conditionally independent if:

$$\boxed{\mathbf{P}(A|B \cap C) = \mathbf{P}(A|C)}$$

Conditional Independence

- Given a (conditioning) event C, the events A and B are called conditionally independent if:

$$\boxed{\mathbf{P}(A \cap B|C) = \mathbf{P}(A|C).\mathbf{P}(B|C)}$$

To derive an alternative characterization of conditional independence, we use the definition of the conditional probability and the multiplication rule, to write

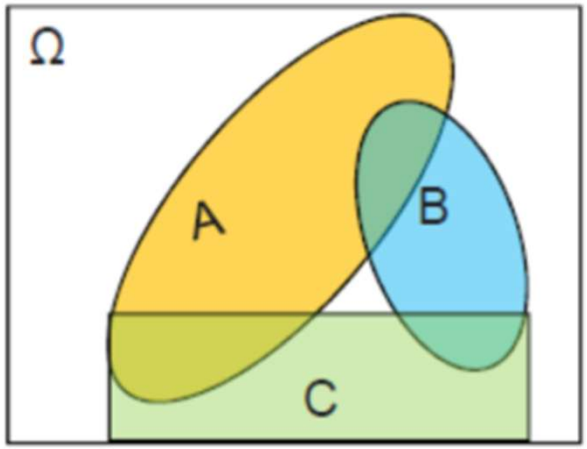
$$\begin{aligned}\mathbf{P}(A \cap B \,|\, C) &= \frac{\mathbf{P}(A \cap B \cap C)}{\mathbf{P}(C)} \\ &= \frac{\mathbf{P}(C)\mathbf{P}(B \,|\, C)\mathbf{P}(A \,|\, B \cap C)}{\mathbf{P}(C)} \\ &= \mathbf{P}(B \,|\, C)\mathbf{P}(A \,|\, B \cap C).\end{aligned}$$

- Given a (conditioning) event C, the events A and B are called conditionally independent if:

$$\boxed{\mathbf{P}(A|B \cap C) = \mathbf{P}(A|C)}$$

Conditioning may Affect Independence!

- Assume A and B are independent
- If we are told event C has occurred, are A and B independent (in this new conditional sample space)?



Conditioning may Affect Independence

Example 1.20. Consider two independent fair coin tosses, in which all four possible outcomes are equally likely. Let

$$H_1 = \{\text{1st toss is a head}\},$$
$$H_2 = \{\text{2nd toss is a head}\},$$
$$D = \{\text{the two tosses have different results}\}.$$

The events H_1 and H_2 are (unconditionally) independent. But

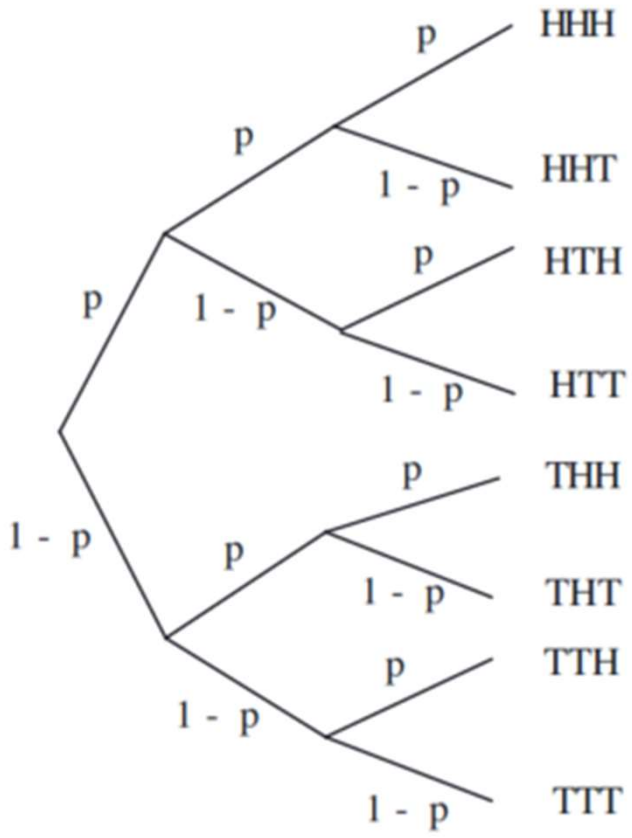
Are H_1 and H_2 conditionally independent with respect to event D.

- 3 tosses of a biased coin:
  $\mathbf{P}(H) = p$, $\mathbf{P}(T) = 1 - p$

$\mathbf{P}(THT) =$

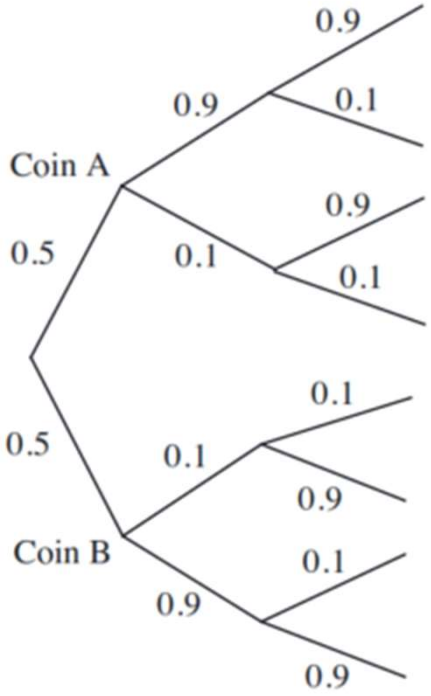
$\mathbf{P}(1 \text{ head}) =$

$\mathbf{P}(\text{first toss is H} \mid 1 \text{ head}) =$



Conditioning may Affect Independence!

- Two unfair coins, A and B:
  $\mathbf{P}(H \mid \text{coin } A) = 0.9$, $\mathbf{P}(H \mid \text{coin } B) = 0.1$
  choose either coin with equal probability
- Once we know it is coin A, are tosses independent?
- If we do not know which coin it is, are tosses independent?
  - Compare:
    $\mathbf{P}(\text{toss } 11 = H)$
    $\mathbf{P}(\text{toss } 11 = H \mid \text{first 10 tosses are heads})$



Independence of a Collection of Events

The definition of independence can be extended to multiple events.

Definition of Independence of Several Events

We say that the events $A_1, A_2, \ldots, A_n$ are **independent** if

$$\mathbf{P}\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} \mathbf{P}(A_i), \qquad \text{for every subset } S \text{ of } \{1, 2, \ldots, n\}.$$

Events $A_1, A_2, \ldots, A_n$
are called **independent** if:

$$\mathrm{P}(A_i \cap A_j \cap \cdots \cap A_q) = \mathrm{P}(A_i)\mathrm{P}(A_j)\cdots\mathrm{P}(A_q)$$

for any distinct indices $i, j, \ldots, q$,
(chosen from $\{1, \ldots, n\}$)

Independence of a Collection of Events

For the case of three events, A_1, A_2, and A_3, independence amounts to satisfying the four conditions

$$\begin{aligned}
\mathbf{P}(A_1 \cap A_2) &= \mathbf{P}(A_1)\,\mathbf{P}(A_2), \\
\mathbf{P}(A_1 \cap A_3) &= \mathbf{P}(A_1)\,\mathbf{P}(A_3), \\
\mathbf{P}(A_2 \cap A_3) &= \mathbf{P}(A_2)\,\mathbf{P}(A_3), \\
\mathbf{P}(A_1 \cap A_2 \cap A_3) &= \mathbf{P}(A_1)\,\mathbf{P}(A_2)\,\mathbf{P}(A_3).
\end{aligned}$$

Pairwise Independence DOES NOT Imply Independence!

Example 1.22. Pairwise Independence does not Imply Independence. Consider two independent fair coin tosses, and the following events:

$$H_1 = \{\text{1st toss is a head}\},$$
$$H_2 = \{\text{2nd toss is a head}\},$$
$$D = \{\text{the two tosses have different results}\}.$$

The events H_1 and H_2 are independent, by definition. To see that H_1 and D are independent, we note that

$$\mathbf{P}(D \mid H_1) = \frac{\mathbf{P}(H_1 \cap D)}{\mathbf{P}(H_1)} = \frac{1/4}{1/2} = \frac{1}{2} = \mathbf{P}(D).$$

Similarly, H_2 and D are independent. On the other hand, we have

$$\mathbf{P}(H_1 \cap H_2 \cap D) = 0 \neq \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \mathbf{P}(H_1)\mathbf{P}(H_2)\mathbf{P}(D),$$

and these three events are not independent.

Is $\mathbf{P}(A \cap B \cap C) = \mathbf{P}(A)\mathbf{P}(B)P(C)$ Enough for Independence of 3 Events A, B, C

Example 1.23. The Equality $\mathbf{P}(A_1 \cap A_2 \cap A_3) = \mathbf{P}(A_1)\,\mathbf{P}(A_2)\,\mathbf{P}(A_3)$ is not Enough for Independence. Consider two independent rolls of a fair six-sided die, and the following events:

$$A = \{\text{1st roll is 1, 2, or 3}\},$$

$$B = \{\text{1st roll is 3, 4, or 5}\},$$

$$C = \{\text{the sum of the two rolls is 9}\}.$$

We have

$$\mathbf{P}(A \cap B) = \frac{1}{6} \neq \frac{1}{2} \cdot \frac{1}{2} = \mathbf{P}(A)\mathbf{P}(B),$$

$$\mathbf{P}(A \cap C) = \frac{1}{36} \neq \frac{1}{2} \cdot \frac{4}{36} = \mathbf{P}(A)\mathbf{P}(C),$$

$$\mathbf{P}(B \cap C) = \frac{1}{12} \neq \frac{1}{2} \cdot \frac{4}{36} = \mathbf{P}(B)\mathbf{P}(C).$$

Thus the three events A, B, and C are not independent, and indeed no two of these events are independent. On the other hand, we have

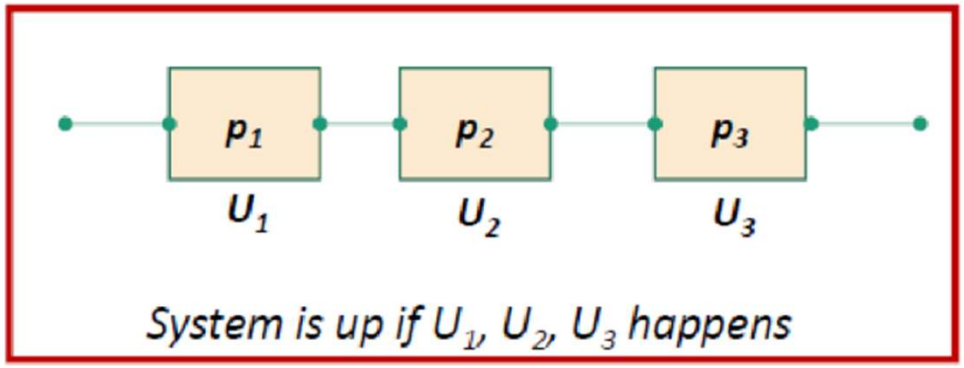
$$\mathbf{P}(A \cap B \cap C) = \frac{1}{36} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{4}{36} = \mathbf{P}(A)\mathbf{P}(B)\mathbf{P}(C).$$

Application of Independence: Reliability

- In probabilistic models of complex systems involving several components, it is often convenient to assume that behaviors of the components are uncoupled/independent. This typically simplifies the calculations and the analysis.
- Systems in Series

Unit i is serviceable w.p. p_i & denoted by U_i

Unit i is unserviceable w.p. $(1 - p_i)$ & denoted by F_i



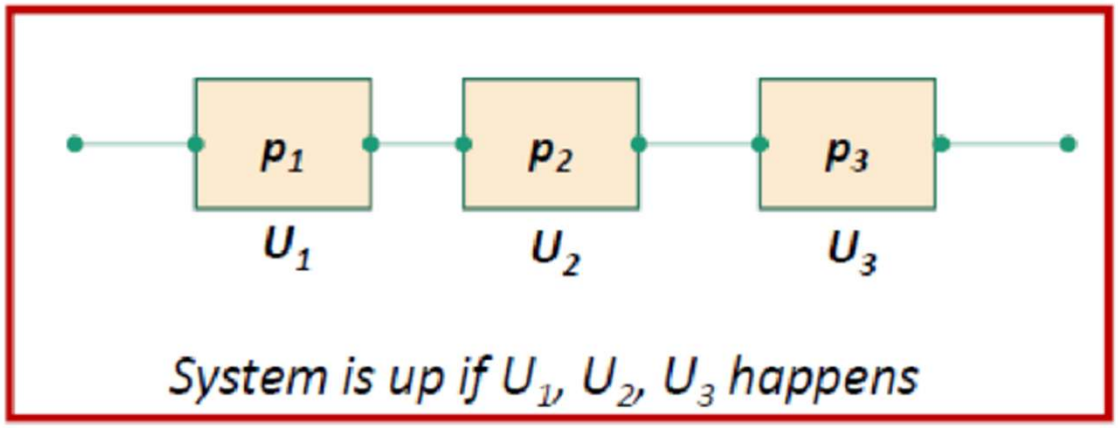
$\mathbf{P}(system\ is\ up) =$

Application of Independence: Reliability

- In probabilistic models of complex systems involving several components, it is often convenient to assume that behaviors of the components are uncoupled/independent. This typically simplifies the calculations and the analysis.
- Systems in Series

Unit i is serviceable w.p. p_i & denoted by U_i

Unit i is unserviceable w.p. $(1 - p_i)$ & denoted by F_i



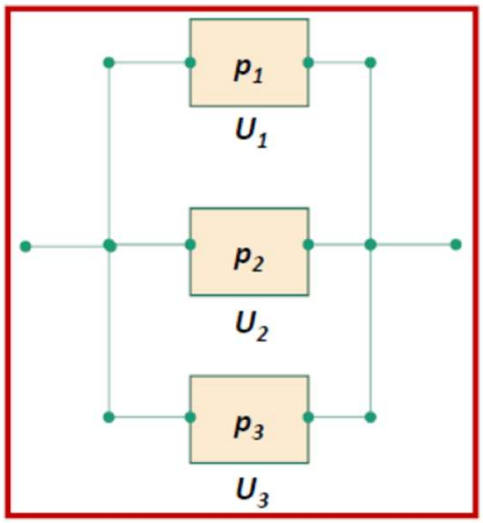
$$\mathbf{P}(system\ is\ up) = \mathbf{P}(U_1 \cap U_2 \cap U_3) = \mathbf{P}(U_1).\mathbf{P}(U_2).\mathbf{P}(U_3) = p_1.p_2.p_3$$

Application of Independence: Reliability

- In probabilistic models of complex systems involving several components, it is often convenient to assume that behaviors of the components are uncoupled/independent. This typically simplifies the calculations and the analysis.
- Systems in Parallel

Unit i is serviceable w.p. p_i & denoted by U_i

Unit i is unserviceable w.p. $(1 - p_i)$ & denoted by F_i



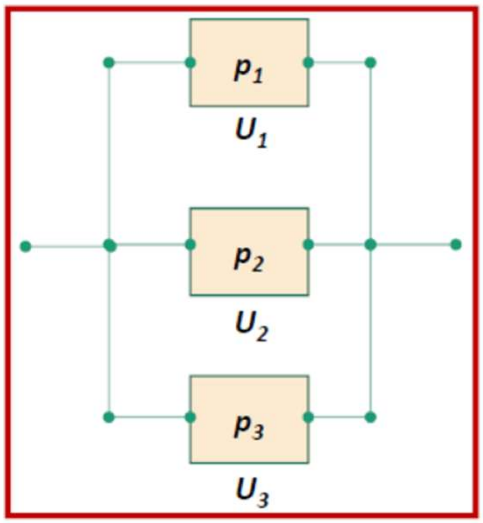
$$\mathbf{P}(system\ is\ up) =$$

Application of Independence: Reliability

- In probabilistic models of complex systems involving several components, it is often convenient to assume that behaviors of the components are uncoupled/independent. This typically simplifies the calculations and the analysis.
- Systems in Parallel

Unit i is serviceable w.p. p_i & denoted by U_i

Unit i is unserviceable w.p. $(1 - p_i)$ & denoted by F_i

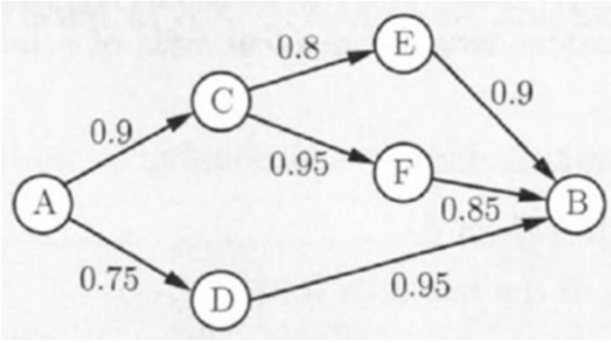


$$\begin{aligned}
\mathbf{P}(system\ is\ up) &= \mathbf{P}(U_1 \cup U_2 \cup U_3) \\
&= 1 - \mathbf{P}(U_1 \cup U_2 \cup U_3)^C \\
&= 1 - \mathbf{P}(U_1^C \cap U_2^C \cap U_3^C) \\
&= 1 - \mathbf{P}(F_1 \cap F_2 \cap F_3) \\
&= 1 - \mathbf{P}(F_1).\mathbf{P}(F_2).\mathbf{P}(F_3) \\
&= 1 - (1 - p_1).(1 - p_2).(1 - p_2)
\end{aligned}$$

Application of Independence: Reliability

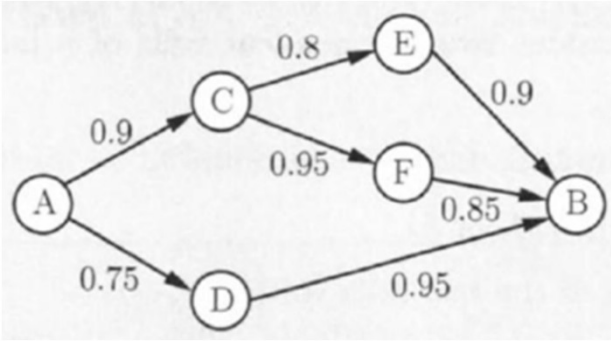
Example 1.24. Network Connectivity. A computer network connects two nodes A and B through intermediate nodes C, D, E, F, as shown in Fig. 1.15(a). For every pair of directly connected nodes, say i and j, there is a given probability p_{ij} that the link from i to j is up. We assume that link failures are independent of each other. What is the probability that there is a path connecting A and B in which all links are up?

$\mathbf{P}(C \rightarrow B)$



Application of Independence: Reliability

Example 1.24. Network Connectivity. A computer network connects two nodes A and B through intermediate nodes C, D, E, F, as shown in Fig. 1.15(a). For every pair of directly connected nodes, say i and j, there is a given probability p_{ij} that the link from i to j is up. We assume that link failures are independent of each other. What is the probability that there is a path connecting A and B in which all links are up?

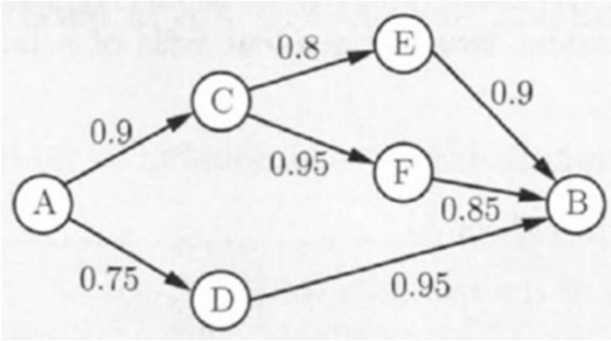


$$
\begin{aligned}
&\mathbf{P}(C \to B) \\
&= 1 - \big(1 - \mathbf{P}(C \to E \text{ and } E \to B)\big)\big(1 - \mathbf{P}(C \to F \text{ and } F \to B)\big) \\
&= 1 - (1 - p_{CE}p_{EB})(1 - p_{CF}p_{FB}) \\
&= 1 - (1 - 0.8 \cdot 0.9)(1 - 0.95 \cdot 0.85) \\
&= 0.946,
\end{aligned}
$$

$$\mathbf{P}(A \to B)$$

Application of Independence: Reliability

Example 1.24. Network Connectivity. A computer network connects two nodes A and B through intermediate nodes C, D, E, F, as shown in Fig. 1.15(a). For every pair of directly connected nodes, say i and j, there is a given probability p_{ij} that the link from i to j is up. We assume that link failures are independent of each other. What is the probability that there is a path connecting A and B in which all links are up?



$$
\begin{aligned}
&\mathbf{P}(C \to B)\\
&= 1 - \big(1 - \mathbf{P}(C \to E \text{ and } E \to B)\big)\big(1 - \mathbf{P}(C \to F \text{ and } F \to B)\big)\\
&= 1 - (1 - p_{CE}p_{EB})(1 - p_{CF}p_{FB})\\
&= 1 - (1 - 0.8 \cdot 0.9)(1 - 0.95 \cdot 0.85)\\
&= 0.946,
\end{aligned}
$$

$$\mathbf{P}(A \to C \text{ and } C \to B) = \mathbf{P}(A \to C)\mathbf{P}(C \to B) = 0.9 \cdot 0.946 = 0.851.$$

$$\mathbf{P}(A \to D \text{ and } D \to B) = \mathbf{P}(A \to D)\mathbf{P}(D \to B) = 0.75 \cdot 0.95 = 0.712,$$

and finally we obtain the desired probability

$$
\begin{aligned}
\mathbf{P}(A \to B) &= 1 - \big(1 - \mathbf{P}(A \to C \text{ and } C \to B)\big)\big(1 - \mathbf{P}(A \to D \text{ and } D \to B)\big)\\
&= 1 - (1 - 0.851)(1 - 0.712)\\
&= 0.957.
\end{aligned}
$$